Learning axial Green surrogates for elliptic problems with variable coefficients

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Abstract Green's function representations express elliptic solutions as convolutions with kernels whose singularity and boundary traces remain difficult for a neural networks to approximate. We introduce *GreenNet*, which integrates a feed-forward network with the closed-form one-dimensional Poisson Green's function so Dirichlet condition and flux jumps are enforced analytically while the network learns only smooth corrections. *CouplingNet* predicts horizontal and vertical flux-divergence components based on those kernels to enforce axial consistency without retraining a multidimensional operator. On high-contrast exponential coefficients, GreenNet achieves orders-of-magnitude lower kernel errors than neural-only surrogates, and applying the same analytic corrective structure within a two-dimensional DeepONet preserves this advantage, indicating a practical path toward reusable axial Green surrogates.

1 Introduction

Consider a one-dimensional variable coefficient elliptic problem

$$\begin{aligned} -\frac{d}{dx}\left(\kappa(x)\frac{du}{dx}(x)\right) &= f(x) \quad \text{for } x \in (a,b), \\ u(a) = g_a, \quad u(b) &= g_b, \end{aligned} \tag{1.1}$$

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where $\kappa(x) > 0$ denotes the coefficient function and f is a forcing term. The Green's function technique generates the solution of (1.1) as an integral operator representation acting on f, which is expressed as a convolution of a Green's kernel and f. When κ varies significantly, closed-form expressions for the Green's kernel are rarely available, making it difficult to use the classical representation directly. Throughout this paper we treat (1.1) as the model solved along each axis; the one-dimensional kernels learned in this setting are later coupled across horizontal and vertical segments to reconstruct higher-dimensional solutions.

Recent data-driven surrogates learn the Green's kernel directly from paired samples of f and u, avoiding the need for closed-form expressions. Boullé et al. [1] combined rational activation functions with symmetry constraints to approximate Green's functions accurately, while GF-Net [2] solves PINN subproblems with localized point sources to learn kernels that generalize across irregular two-dimensional domains. Boundary-integral networks such as BI-GreenNet [3] combine fundamental solutions with learned boundary representations so they can treat complex geometries, variable coefficients, and even unbounded exterior domains, whereas Neural Green's Functions [4] generate geometry-aware spectral decompositions that transfer across many shapes with efficient inference. These advances emphasize the benefits of respecting the Green representation, but they still rely on heavy multi-dimensional training, expensive boundary solves, or geometry encoders to ensure boundary traces and flux jumps. Motivated by these developments, we design a surrogate that (i) preserves the analytical structure of variable-coefficient kernels and (ii) can be reused when coupling axial segments in higher dimensions without retraining an operator in the full domain.

Our contributions are (i) GreenNet, which couples the closed-form Poisson Green's function and its antiderivative with a neural network so boundary traces and flux jumps are enforced analytically while the network learns only smooth corrections, (ii) CouplingNet, which reuses those kernels to predict horizontal and vertical flux-divergence components and enforce the axial consistency constraint, and (iii) numerical evidence that embedding the analytical corrector yields lower errors than neural-only surrogates, even inside a two-dimensional DeepONet wrapper.

Section 2 reviews the one-dimensional Green representation, Section 3 details *GreenNet* and outlines *CouplingNet*, and Section 4 reports numerical experiments for the one-dimensional GreenNet learning task and its reuse inside a two-dimensional DeepONet. We conclude by outlining how CouplingNet validation will extend the study.

2 One-Dimensional Green Representations

For problem (1.1), we define the Green's function $G(x, \xi)$ as the solution to

$$
\begin{aligned}
&-\frac{d}{dx}\left(\kappa(x)\frac{dG}{dx}(x,\xi)\right) = \delta(x-\xi),\\
&G(a,\xi) = G(b,\xi) = 0,
\end{aligned}
\tag{2.1}
$$

with the derivative jump $\kappa(\xi)\left[\partial_x G(\xi^+,\xi) - \partial_x G(\xi^-,\xi)\right] = 1$. For homogeneous Dirichlet data the solution admits the representation

$$
u(x) = \int_a^b G(x,\xi) f(\xi)\, d\xi, \tag{2.2}
$$

whereas inhomogeneous boundary data are incorporated by adding the unique affine interpolant that matches $u(a) = g_a$ and $u(b) = g_b$. Once G is known, evaluating (2.2) produces the solution for any source f without solving an additional boundary value problem.

The combination of (1.1), (2.1) and (2.2) shows that learning the kernel is equivalent to learning the solution operator. Consequently, accurately approximating G as an explicit integral kernel provides computational advantages of neural operator models.

3 Neural Approximation of One-Dimensional Kernels

3.1 Relation to data-driven discovery

The framework of Boullè et al. [1] learns Green's kernels by leveraging neural architectures that employ rational activation functions, ensuring boundary conditions and singular structures are explicitly enforced. Their results emphasize the importance of exactly capturing boundary traces and singular features. We follow this principle and adapt the architecture to one-dimensional elliptic operators with variable coefficient.

3.2 GreenNet architecture

For the sake of convenience, we consider (2.1) on the interval $(0,1)$. GreenNet uses a fully connected network enhanced with analytical primitives that enforce boundary values and flux jump. Let G_P denote the Green's function for the Poisson equation on $(0,1)$ with homogeneous Dirichlet boundary conditions, then it is expressed as

$$
G_P(x,\xi) = \begin{cases} x(1-\xi), & x < \xi,\\ \xi(1-x), & x \geq \xi,\end{cases}
$$

with its antiderivative

$$H_P(x,\xi) = \int_0^x G_P(s,\xi)\,ds = \begin{cases} \frac{1}{2}x^2(1-\xi), & x < \xi, \\ \frac{1}{2}\xi(2x - x^2 - \xi), & x \geq \xi. \end{cases}$$

We propose the GreenNet kernel

$$\begin{aligned} G(x,\xi) = &\left[x\xi(1-\xi)\left((1-x)\widehat{G}_{\mathrm{NN}}(x,\xi) - \tfrac{1}{2}x\right) + H_P(x,\xi)\right]\frac{\kappa'(\xi)}{\kappa(\xi)\kappa(x)} \\ &+ \frac{1}{\kappa(x)}\,G_P(x,\xi), \end{aligned} \tag{3.1}$$

where $\widehat{G}_{\mathrm{NN}}$ is the kernel from a feedforward network. The coefficients multiplied to G_P and H_P in (3.1) are selected to reproduce the singular feature generated by $L = -\partial_x(\kappa(x)\partial_x)$. Applying L to $\frac{1}{\kappa(x)}G_P(x,\xi)$ delivers the Dirac delta, while the term proportional to $\frac{\kappa'(\xi)}{\kappa(\xi)\kappa(x)}H_P(x,\xi)$ cancels the Heaviside contributions that arise when the operator L differentiates H_P twice. Consequently, the result of L acting on the residual component $x(1-x)\xi(1-\xi)\widehat{G}_{\mathrm{NN}}(x,\xi)$ is continuous, so the neural network learns only smooth corrections rather than distributional jumps, enforcing the boundary behavior and flux discontinuity exactly. The multiplicative gate $x(1-x)\xi(1-\xi)$ forces the residual to vanish at the boundaries, and subtracting $\frac{1}{2}x$ removes the nonzero trace of H_P at $x = 1$.

Given paired samples $\{(f_i, u_i)\}_{i=1}^N$ on a uniform grid, GreenNet minimizes the reconstruction loss

$$\mathcal{L}_{\mathrm{Green}} = \frac{1}{N}\sum_{i=1}^{N}\left\| u_i - \int_0^1 G_\theta(x,\xi) f_i(\xi)\,d\xi \right\|_{L^2}^2,$$

with Simpson's rule evaluating the integral. In practice, we implement $\widehat{G}_{\mathrm{NN}}$ as a fully connected network consisting of two hidden layers with width 256 and rational activations following Boullé et al. [1].

4 Extension to Higher-Dimensional Problems

4.1 Challenges for multidimensional kernels

We consider a two-dimensional elliptic problem

$$-\nabla\cdot(\kappa(\mathbf{x})\nabla u(\mathbf{x})) = f(\mathbf{x}), \qquad \Omega \subset \mathbb{R}^2 \tag{4.1}$$

with Dirichlet condition and coefficient $\kappa(\mathbf{x})$, which allows a Green's function representation analogous to (2.2). However, the two-dimensional Green's kernel depends on the domain geometry (Ω) and the coefficient (κ). Because their influences are complicated, the kernel cannot usually be written as an explicit analytical formula;

instead, it must be computed numerically or approximated. Next, we review how to solve (4.1) by decomposing it into one-dimensional problems along the axis-parallel lines and combining the corresponding one-dimensional solutions.

4.2 Axial decomposition with CouplingNet

First, for each interior point of Ω, we consider segments parallel to the axis passing through that point. Let $(\bar{x}, \bar{y})$ be an interior point of Ω. Along a horizontal segment $X_{\bar{y}}$ and a vertical segment $Y_{\bar{x}}$ the solution of (4.1) satisfies

$$\begin{aligned} \partial_x \big(\kappa(x, \bar{y})\, \partial_x u(x, \bar{y})\big) &= \varphi(x, \bar{y}), \\ \partial_y \big(\kappa(\bar{x}, y)\, \partial_y u(\bar{x}, y)\big) &= \psi(\bar{x}, y), \end{aligned} \tag{4.2}$$

where φ and ψ satisfy $\varphi(\bar{x}, \bar{y}) + \psi(\bar{x}, \bar{y}) = f(\bar{x}, \bar{y})$. Kim et al. [5] showed that these axial components must satisfy the coupled integral relation

$$\int_{x_-}^{x_+} G_x(\bar{x}; \xi)\varphi(\xi, \bar{y})\, d\xi + \int_{y_-}^{y_+} G_y(\bar{y}; \eta)\varphi(\bar{x}, \eta)\, d\eta = \int_{y_-}^{y_+} G_y(\bar{y}; \eta) f(\bar{x}, \eta)\, d\eta,$$

where $G_x(\bar{x}; \xi)$ and $G_y(\bar{y}; \eta)$ are 1-D Green's functions on the segments parallel to the x-axis and y-axis crossing $(\bar{x}, \bar{y})$, respectively. In [5], this coupled integral solution produced a linear system to find φ or ψ. This is called the Axial Green's Function Method. In this work, we instead propose *CouplingNet* as a neural component that approximates φ and ψ by leveraging previously learned GreenNet kernels; given samples of κ and f along an axial segment, it produces the corresponding flux divergence. We train the network with a loss that enforces the axial flux-divergence balance relations in (4.2). On horizontal lines, the network predicts $\varphi_\theta(\xi, \bar{y})$, while vertical evaluations satisfy $\psi_\theta(\bar{x}, \eta) = f(\bar{x}, \eta) - \varphi_\theta(\bar{x}, \eta)$. The flux-divergence balance loss measures the residual of the identity $\varphi + \psi = f$ on the intersection set $\mathcal{B}$, while the consistency term compares reconstructions formed with the current GreenNet surrogate G_θ:

$$\mathcal{L}_{\text{FDB}} = \frac{1}{N} \sum_{j=1}^{N} \big\| \varphi_{\theta,j} + \psi_{\theta,j} - f_j \big\|_{L^2(\mathcal{B})}^2,$$

$$\mathcal{L}_{\text{consistency}} = \frac{1}{N} \sum_{j=1}^{N} \big\| u_{\theta,j}^{(x)} - u_{\theta,j}^{(y)} \big\|_{L^2(\mathcal{B})}^2,$$

where $u_{\theta,j}^{(x)}$ and $u_{\theta,j}^{(y)}$ are solutions obtained using Green's function G_θ along the horizontal and vertical segments, respectively. The overall objective $\mathcal{L} = \lambda_1 \mathcal{L}_{\text{FDB}} + \lambda_2 \mathcal{L}_{\text{consistency}}$ balances the two terms. We postpone empirical verification to future work.

5 Numerical Experiments

5.1 A 1-D problem with high contrast coefficient function

We first consider (1.1) with $\kappa(x) = 5\exp(x) - 4.9$ on the interval $(0, 1)$. This choice yields a high-contrast coefficient, providing a challenging benchmark for GreenNet. Training functions are sampled from the Gaussian process $u \sim \mathcal{GP}(0, k_\ell)$ with a squared exponential kernel

$$k_\ell(x, x') = \exp\Big(-\frac{(x - x')^2}{2\ell^2}\Big),$$

and each sample is a random sample of scale length ℓ drawn uniformly from $[0.05, 0.25)$. Samples are evaluated on a uniform grid with mesh size $1/256$ and adjusted to satisfy $u(0) = u(1) = 0$ by subtracting the affine interpolant matching the sampled endpoints. We impose homogeneous Dirichlet conditions so that the solution is given exclusively by the Green's kernel integral; for nonzero boundary values, we can add analytical corrections that do not affect the kernel approximation and do not complicate the training targets. We extract 100 functions for training and 20 functions for validation. Analytic differentiation of the function generated by the squared exponential Gaussian process yields the exact right-hand sides through $f_i = -\frac{\partial}{\partial x}(\kappa \frac{\partial}{\partial x} u_i)$.

These draws therefore cover both highly oscillatory inputs (small ℓ) and slowly varying inputs (large ℓ) so that the learned kernels experience a wide range of forcing patterns [7].

We compare GreenNet with a neural-network-only multilayer perceptron that shares the same width, depth, and optimization schedule but omits the analytical corrector. Figure 1 summarizes the loss histories and kernel errors: GreenNet drives both metrics near 10^{-8} and 10^{-4}, respectively, whereas the comparison network stays around 10^{-4} and 10^{-2}.

Figure 2 visualizes the learned kernels: GreenNet tracks the singular feature and keeps absolute errors below 5×10^{-3}, while the neural-network-only surrogate smooths the interior and accumulates boundary errors near 7×10^{-2}.

5.2 Two-dimensional extension with DeepONet

To demonstrate the reuse of one-dimensional GreenNet kernels in higher dimensions, we consider a 2-D problem on $\Omega = (0, 1)^2$ with $\kappa(x, y) = 5\exp(x + y) - 4.9$. We train DeepONet [6] with a branch–trunk output acting as $\widehat{G}_{\mathrm{NN}}$ in the GreenNet architecture and compare the results to those of the neural-network-only DeepONet obtained by removing the analytical corrector from GreenNet without changing the hyperparameters. Training traces are drawn from the same Gaussian process but restricted to 30 axis-parallel lines (mesh size $1/16$), with 50 training and 10 validation

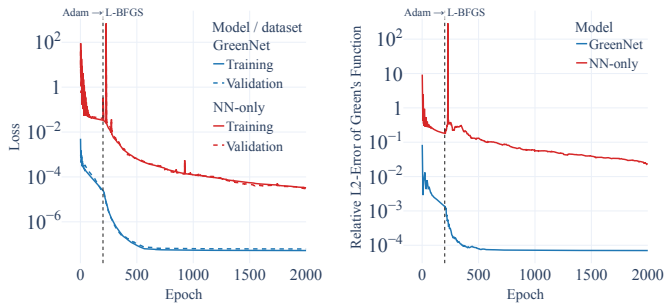

Fig. 1 Training and validation losses (left) and relative L^2 kernel error (right) for GreenNet (blue) and the neural-network-only surrogate (red) on $\kappa(x) = 5\exp(x) - 4.9$. The dashed vertical line marks the switch of optimizers from Adam to L-BFGS.

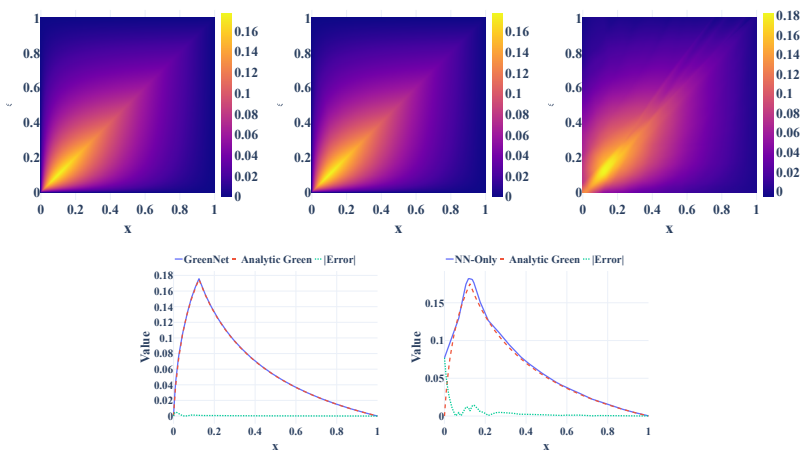

Fig. 2 Green's kernels for $\kappa(x) = 5\exp(x) - 4.9$. Top row: exact Green's function (left), GreenNet surrogate (center), and neural-network-only surrogate (right). Bottom row: vertical sections at $\xi = 0.125$ comparing each surrogate with the exact Green's function (left: GreenNet, right: neural-network-only).

functions per line. Figure 3 shows that embedding GreenNet inside DeepONet keeps losses near 10^{-5} and kernel errors near 10^{-4}, whereas the neural-network-only model remains near 1, indicating that the analytical corrector still matters inside the operator learning architecture.

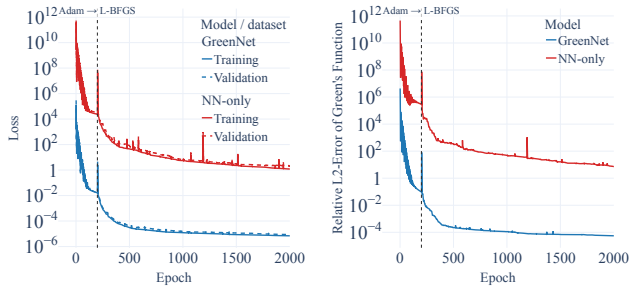

Fig. 3 Loss histories (left) and relative L^2 kernel errors (right) for the two-dimensional DeepONet experiment with $\kappa(x, y) = 5\exp(x + y) - 4.9$; GreenNet (blue) is compared with the neural-network-only counterpart (red).

6 Conclusion

GreenNet couples closed-form Poisson primitives with a neural network so singular features and boundary values are enforced analytically, leaving the network to learn smooth corrections. On the high-contrast one-dimensional benchmark it delivers more than two orders-of-magnitude lower kernel errors than a neural-network-only surrogate, and embedding the learned kernel inside a two-dimensional DeepONet model retains that advantage. Future work will validate the proposed CouplingNet mechanism for coupling the one-dimensional kernels across axes.

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